Grace Nina Smith

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scan for clickable links!

Education

Massachusetts Institute of Technology — **Sep. 2019 – Sep. 2026**
Candidate for Bachelors of Science: Mechanical Engineering — *Cambridge, MA*

Magnolia West High School — **Aug. 2015 – May 2019**
AP Scholar with Distinction, 15 AP Courses | GPA: 4.29, weighted — *Magnolia, TX*

Relevant Coursework

- Medical Device Design
- Nonlinear Dynamics
- Measurement and Instrumentation
- Thermodynamics of Biomolecular Systems
- Digital Instrument Design
- Electronics for Mechanical Systems
- Numerical Analysis
- Social Entrepreneurship

Research Experience

Cell Growth: Surface Area to Volume Scaling — **October 2022 – December 2023**
Mentored by Teemu Miettinen | MIT Koch Institute — *Cambridge, MA*

- Developed a MATLAB automation for the analysis of cell cortex thickness from fluorescent microscopy images.
- Culminated in a paper published in *Current Biology*

Work Experience

3D Design and Fabrication Consultant — **January 2024 - August 2024**
SpectroArt — *Remote*

- Diagnosed and remediated geometric defects in client-produced 3D models, then systematized the process into documented workflows for sustained independent use
- Coached founder through design sessions to develop SLA-specific manufacturing-aware modeling practices in Blender
- Conducted supplier landscape research and qualified vendors for product components, leveraging prior experience in direct manufacturer sourcing

Design Engineering Intern — **Fall 2023**
Transaera — *Somerville, MA*

- Prototyped a portable AC unit with $\geq$30% energy savings
- Introduced engineering team to modern 3D-printing technologies and techniques, saving weeks on lead times for previously-outsourced parts
- Regularly tapped by engineers across the team to workshop novel fabrication and design problems

Mechanical Engineering Tutor — **Summer 2023**
MIT Women's Technology Program | *MIT Mechanical Engineering* — *Cambridge, MA*

- Delivered a rigorous, intensive introduction to engineering to 50 high-school students
- Led engaging classes in brainstorming, design, and rapid-prototyping/manufacturing
- Returned in subsequent years to assist and advise in manufacturing-related activities

Software Engineer — **October 2020 - January 2021**
Uzephra — *Remote*

- Developed a COVID-19 risk model for disease spread on airplanes based on the Wells-Riley model and extended to consider cabin-to-cabin airflow as a factor in virus spread
- Integrated model within architecture of travel web application as weighted contribution to overall demographic-based risk calculation

Technical Skills

Programming Languages and Frameworks: Python, MATLAB, C/C++, Arduino, Julia, MAX/MSP, Java, R, PyTorch, Flask, comfortable in Linux-based systems
Languages: English (native), Russian (native), Spanish (basic/conversational), Japanese (basic/conversational)
Other: Manual and digital fabrication, Fusion360, SOLIDWORKS, KiCad, Adobe Creative Suite, Blender, Railway/Docker